

FUNCTIONS OF SALIVA

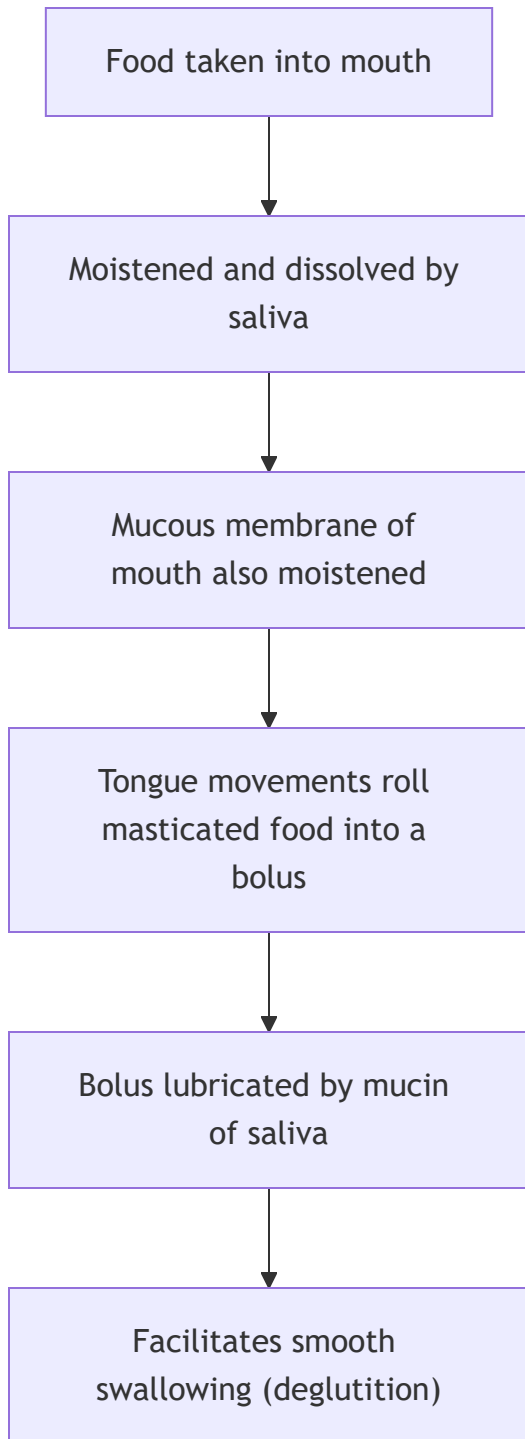
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- Saliva is an essential digestive juice involved in physical, chemical, defensive, and excretory functions.

1. PREPARATION OF FOOD FOR SWALLOWING



2. APPRECIATION OF TASTE

- Taste is a chemical sensation.
- Mechanism :—

Saliva dissolves solid food substances via solvent action



Dissolved substances stimulate taste receptors on taste buds



Enables brain to recognize taste

3. DIGESTIVE FUNCTIONS

- Saliva contains **3 digestive enzymes**:
 - a. Salivary amylase (Ptyalin)
 - b. Maltase
 - c. Lingual lipase

4. CLEANSING AND PROTECTIVE FUNCTIONS

- a. **Mechanical Washing** : Constant secretion of saliva rinses the mouth and teeth, keeping them free of food debris, shed epithelial cells, and foreign particles → **prevents bacterial growth**.
- b. **Lysozyme** : Kills bacteria such as *Staphylococcus*, *Streptococcus*, and *Brucella*.
- c. **Proline-Rich Proteins** : Possess antimicrobial properties and **neutralize toxic substances** (e.g., *tannins*).
- d. **Lactoferrin** : Possesses antimicrobial / iron-chelating properties.
- e. **Enamel Protection** : Proline-rich proteins and lactoferrin protect teeth by **stimulating enamel formation**.
- f. **Immunoglobulin A (IgA)** : Provides **antibacterial and antiviral actions**.
- g. **Mucin** : Protects the mouth by **lubricating the oral mucous membrane**.

5. ROLE IN SPEECH

- Moistening and lubricating the soft parts of the mouth, tongue, and lips with saliva **facilitates speech articulation**.

6. EXCRETORY FUNCTION

- a. **Normal Excretory Products** : Excretes substances like **mercury, potassium iodide, lead, and thiocyanate**.
- b. **Viral Excretion** : Excretes certain viruses (*e.g., Rabies virus, Mumps virus*).
- c. **Pathological Excretion** :
 - **Glucose** : In Diabetes Mellitus
 - **Excess Urea** : In Nephritis
 - **Excess Ca^{2+}** : In Hyperparathyroidism

7. REGULATION OF BODY TEMPERATURE

- **In Animals (Dogs & Cattle)** : Excessive dripping of saliva during **panting** helps in heat loss and body temperature regulation.
- **In Humans** : Sweat glands play the main role in temperature regulation; **saliva does not play a significant role**.

8. REGULATION OF WATER BALANCE

